ETHICAL HACKING SYLLABUS

A comprehensive course outline

Course Overview

This syllabus outlines the curriculum for an Ethical Hacking course designed to provide students with a comprehensive understanding of cybersecurity principles, hacking techniques, and methods for securing systems and networks. The course covers a wide range of topics, from basic Linux commands to advanced red teaming strategies. Students will engage in practical exercises and real-world scenarios to develop their skills and prepare them for careers in cybersecurity.

Daily Schedule

Day 1: Introduction to Ethical Hacking

- Definition and purpose of ethical hacking
- Difference between ethical hacking and malicious hacking
- Legal aspects, scope, and responsible disclosure
- Types of hackers and common attack surfaces
- Ethical hacking process and methodology

Day 2: Lab Setup & Termux Introduction

- Introduction to virtual machines
- Setting up Kali Linux in a virtual environment
- Installing and configuring essential tools
- Network configuration basics
- Basic Linux commands for hackers
- Introduction to Android Termux (installation, storage permissions, repositories)
- Learning basic Linux commands using Termux on Android

Day 3: Hacking Tools

- Overview of essential hacking and security tools
- Introduction to Nmap, Metasploit, Wireshark, Burp Suite
- Installation and configuration of tools
- Basic usage examples and guided exercises

Day 4: Social Media Hacking and Prevention

- Common social media attacks and real-world scenarios
- Phishing and social engineering on social platforms
- Protecting personal and professional accounts
- Practical exercises on securing social media accounts

Day 5: Password Hacking

- Types of password attacks (brute force, dictionary, rainbow tables)
- Tools for password hacking (John the Ripper, Hashcat)
- Password policies and techniques for creating strong passwords
- Practical lab: Cracking password hashes in a controlled environment

Day 6: Cryptography

- Basics of cryptography (symmetric and asymmetric)
- Common encryption algorithms and real-world use cases
- Encryption tools (GPG, VeraCrypt)
- Hands-on: Encrypting and decrypting data

Day 7: Location, Camera Hacking, and Prevention

- Techniques for geolocation tracking and camera hacking
- Tools and methods used for such attacks
- Best practices to protect devices from these attacks
- Practical exercises on securing mobile and laptop devices

Day 8: Social Engineering & Phishing

- Introduction to social engineering concepts
- Types of social engineering attacks (pretexting, baiting, tailgating, vishing, smishing)
- Social engineering and phishing campaigns (email, SMS, social media)
- Psychological aspects of manipulation and human factors
- Case studies, role-play scenarios, and prevention techniques

Day 9: Red Teaming Tools

- Introduction to red teaming and adversary emulation
- Essential red teaming tools and frameworks
- Scenario-based exercises and planning an engagement
- Practical lab: Simulating a basic red team attack in a lab setup

Day 10: Securing From Hackers

- Best practices for securing systems and networks
- Security frameworks and standards (NIST, ISO 27001 – overview)
- Implementing basic security policies and procedures
- Practical exercises on hardening and securing a small network

Day 11: Dark Web

- Understanding the deep web vs dark web
- Accessing the dark web (Tor, I2P) safely and legally
- Common activities on the dark web
- Legal and ethical considerations for security professionals

Day 12: Anonymity & Privacy (“HIDE FROM SOCIAL WORLD”)

- Techniques for maintaining anonymity and privacy online
- Tools for anonymous browsing and communication
- Practical lab: Using VPNs, Tor, secure email, and privacy-focused browsers
- Personal digital hygiene and footprint minimization

Day 13: OSINT (OPEN-SOURCE INTELLIGENCE)

- Introduction to OSINT and its role in ethical hacking and investigations
- OSINT collection from social media, public records, and technical sources
- Basic OSINT tools and frameworks
- Hands-on: Conducting a guided OSINT investigation with defined scope and ethics

Day 14: TERMUX REMOTE ACCESS & SSH

- Understanding SSH and secure remote access concepts
- Setting up and configuring SSH in Termux
- Connecting from Termux to remote Linux servers safely
- File transfer and remote command execution from mobile devices

Day 15: LINUX FROM MEDIUM TO A

- Understanding Advance linux tools

Next Steps

Following this syllabus, participants will have gained a solid foundation in ethical hacking. To further enhance your skills:

- Continue practicing the techniques learned in the course.
- Explore advanced cybersecurity topics and certifications.
- Engage with the cybersecurity community to stay updated on the latest trends and threats.